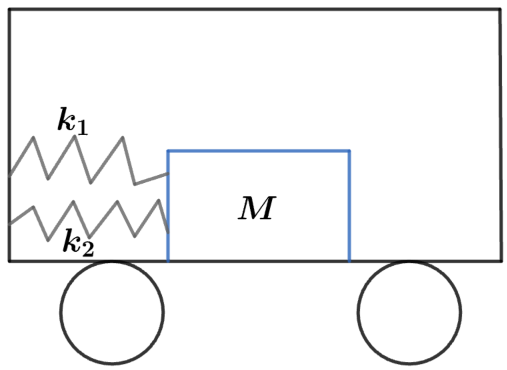
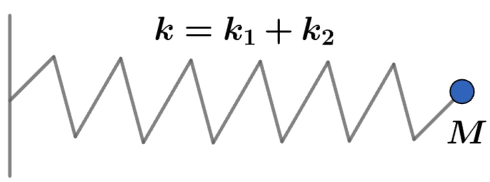
2012 F=ma Exam: Problem 16

Kevin S. Huang

The two springs are connected in parallel since both are connected to the block on one end and the cart on the other end. We can move the second spring to the same side as the first to make this clear:



Spring constants in parallel add so the setup is equivalent to a mass-spring system with $k = k_1 + k_2$.



Recall the frequency of a mass-spring system is given by

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{M}} = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{M}}$$

The frequency is unaffected by acceleration or gravity which only shift the equilibrium position. Thus, the answer is $\boxed{\text{E}}$.